

Corrigendum

Theorem 2 of “Markov-based continuous learning with diversion of data distribution direction for streaming data in limited memory”

Peemapat Wongsriphisan Kitiporn Plaimas Chidchanok Lursinsap

Expert Systems With Applications, 298 (2026) 129818

DOI: [10.1016/j.eswa.2025.129818](https://doi.org/10.1016/j.eswa.2025.129818)

This notice addresses a mathematical error in Theorem 2 of the originally published paper and provides the corrected theoretical mechanism underlying the D^4 algorithm. **This theoretical correction does not affect the algorithm’s implementation, functionality, or the experimental results reported in the paper.**

1 Correction to Theorem 2

In the original paper, the proof for Theorem 2 contains an algebraic sign error in the final steps. When taking the reciprocal and subtracting from a constant, the inequality sign must be reversed. Consequently, the original conclusion stating that the distance between the projected hyper-ellipsoids increases ($d(\hat{\Omega}_a, \hat{\Omega}_b) \geq d(\Omega_a, \Omega_b)$) is incorrect.

The mathematically corrected conclusion is the opposite: the projected distance is always less than or equal to the original distance,

$$d(\hat{\Omega}_a, \hat{\Omega}_b) \leq d(\Omega_a, \Omega_b). \quad (10, \text{corrected})$$

Therefore, the resolution of classification ambiguity is not achieved by proving the hyper-ellipsoids become further apart in terms of Definition 2, but by substituting the original, unreliable classification scores with a new set of projected scores that exhibit greater separability.

2 Corrected Mechanism: Differential Penalty

The theoretical justification for D^4 relies on the asymmetric reduction of the metric scores. Let $\psi_a(\mathbf{x})$ be the original metric for a hyper-ellipsoid Ω_a , and $\psi'_a(\mathbf{x})$ be the projected metric after selecting a subset of eigenvector indices D_a .

The original metric can be partitioned into the projected metric and a non-negative *removed penalty* term $\Delta_a(\mathbf{x})$ representing the summation over the discarded axes (D'_a):

$$\Delta_a(\mathbf{x}) = \sum_{j \in D'_a} \frac{((\mathbf{x} - \mathbf{c}_a)^T \mathbf{u}_{a,j})^2}{w_{a,j}^2}$$

Since $w_{a,j}^2 > 0$, it is guaranteed that $\Delta_a(\mathbf{x}) \geq 0$. The relationship between the scores simplifies to:

$$\psi_a(\mathbf{x}) = \psi'_a(\mathbf{x}) + \Delta_a(\mathbf{x})$$

It follows directly that the projected score is always less than or equal to the original score ($\psi'_a(\mathbf{x}) \leq \psi_a(\mathbf{x})$).

The effectiveness of D^4 arises because this reduction is **not uniform** for both classes. In the overlap region (where $\psi_a < 0$ and $\psi_b < 0$), the algorithm substitutes the ambiguous decision $\arg \min(\psi_a, \psi_b)$ with a more robust decision:

$$\arg \min(\psi'_a, \psi'_b) \equiv \arg \min(\psi_a - \Delta_a, \psi_b - \Delta_b)$$

A central design goal of D^4 is to enhance the score separability, $|\psi'_a(\mathbf{x}) - \psi'_b(\mathbf{x})|$. This is achieved when the *differential penalty* $|\Delta_a(\mathbf{x}) - \Delta_b(\mathbf{x})|$ amplifies the original separation.

The D^4 **Compactness criterion** serves as the primary heuristic for achieving favorable differential penalties. By prioritizing the selection of eigenvectors corresponding to the smallest eigenvalues, it guarantees that the discarded axes (D'_a) are those with the largest widths ($w_{a,j}^2$). Maximizing the denominators in the penalty term systematically reduces the per-unit-projection contribution of each removed axis. This geometric criterion exploits the natural spread of the classes to create asymmetric penalties, transforming an ambiguous classification into a decisive one.

3 Impact Summary

Component	Status	Reason
Algorithm implementation	Unaffected	Theorem 2 concerns analysis only
Experimental results	Unaffected	No algorithmic change
Theorem 1 (Parallelism)	Valid	Independent proof
Theorem 3 (Closeness)	Valid	Uses triangle inequality only
Theorem 2 (Eq. 10)	Replaced	See Section 2

The corrected theoretical framework provides a more precise characterization of the D^4 mechanism. All conclusions drawn from the experimental evaluation in the original paper remain valid.